

Human-Guided Augmentation Filtering for Data-Scarce Image Classifiers: Gains and Robustness Limits

Abstract—Shortcut learning can cause image classifiers to rely on spurious background cues rather than class-relevant features, reducing reliability under distribution shifts. Adversarial attacks represent a distinct threat that can similarly expose reliance on brittle features. This paper proposes a human-guided data augmentation framework that combines pretrained foreground localization, foreground-focused candidate augmentation, and semantic quality filtering to construct more reliable training samples in limited-data settings. To examine whether human review provides benefits beyond automated filtering, we compare three training conditions: unfiltered augmentation, confidence-based automated filtering, and human-supervised filtering. Within each dataset, all conditions use matched candidate counts and the same dual-objective classification–reconstruction architecture. We evaluate the framework on MNIST, CIFAR-10, and STL-10 under clean testing and white-box FGSM and PGD attacks. Human-supervised filtering achieves the highest clean classification accuracy with the custom architecture across all three datasets, improving over the unfiltered baseline by 0.20, 7.00, and 3.63 percentage points on MNIST, CIFAR-10, and STL-10, respectively. Its adversarial benefit, however, is not consistent. Human filtering outperforms automated filtering under moderate FGSM perturbations on MNIST, whereas confidence-based filtering generally matches or exceeds it under FGSM on CIFAR-10 and STL-10. Under stronger PGD attacks, all conditions degrade substantially. These results indicate that human curation can improve clean classification performance, but its adversarial benefit varies across datasets and perturbation strengths. The proposed framework is therefore a data-centric training strategy rather than a substitute for dedicated adversarial training.

Index Terms—Shortcut learning, adversarial robustness, human-in-the-loop learning, foreground-aware augmentation, image segmentation, data augmentation, FGSM, PGD

I. INTRODUCTION

Deep neural networks may achieve high predictive accuracy by exploiting statistical patterns that correlate with class labels but are not causally related to the task. This behavior, known as *shortcut learning*, can cause models to rely on background textures, colors, or other incidental cues rather than semantically meaningful features [1]. Such reliance may remain hidden under standard testing but substantially reduce reliability under distribution shifts.

Adversarial vulnerability is distinct from shortcut learning, although both may involve brittle or non-robust features. Prior work has explored supervised, semi-supervised, self-supervised, and contrastive strategies for improving adversarial robustness [2]–[7]. Robust few-shot learning has also been investigated for limited-data settings [8]. However, many of

these approaches require adversarial-example generation, specialized pretraining, or additional optimization stages.

Data augmentation offers a more efficient way to improve representation diversity and generalization. Foreground-aware methods such as ObjectAug show that applying transformations to localized objects can produce more focused training samples [9], while segmentation models such as Mask2Former enable reliable foreground localization across visual domains [10]. Nevertheless, conventional augmentation pipelines typically accept all transformed samples without verifying whether they preserve the source image’s semantic identity. Aggressive transformations may distort class-defining features, introduce artifacts, or generate ambiguous samples.

This work investigates whether human-guided filtering of foreground-focused augmentations can improve clean generalization and adversarial robustness in limited-data image classification. The proposed framework suppresses distracting background content, generates multiple augmented views, and applies a human quality-control gate to retain semantically valid samples. The selected candidates are used to train a dual-objective classification–reconstruction model that encourages identity preservation across transformations [11].

To determine whether human judgment provides benefits beyond filtering alone, we compare three conditions: unfiltered augmentation, confidence-based automated filtering, and human-supervised filtering. The framework is evaluated on MNIST, CIFAR-10, and STL-10 under clean testing and white-box FGSM and PGD attacks. We also examine its behavior on CIFAR-10 and STL-10 with four pretrained encoders: MobileNetV2, ResNet50, VGG16, and VGG19. This is because robustness may vary across architectures and transfer-learning configurations [12], [13].

The principal contributions are as follows:

- We propose a human-guided augmentation framework combining foreground localization, systematic augmentation, semantic filtering, and dual-objective classification–reconstruction learning.
- We conduct a three-way comparison among unfiltered, automatically filtered, and human-supervised augmentation to isolate the benefit of human judgment.
- We evaluate the framework on MNIST, CIFAR-10, and STL-10, showing that human filtering consistently achieves the highest clean accuracy with the custom architecture.

- We identify a dataset-dependent robustness trade-off: human filtering performs better under moderate-to-high FGSM perturbations on MNIST, whereas automated filtering generally matches or exceeds it on CIFAR-10 and STL-10.

The remainder of this paper reviews related work in Section II, presents the proposed methodology and experimental setup in Sections III and IV, respectively. Next, it discusses the results in Section V, and concludes with limitations and future directions in Section VI.

II. RELATED WORK

This work intersects shortcut mitigation, foreground-aware augmentation, adversarially robust representation learning, and human-guided data curation. Prior studies typically address these areas separately; we examine whether semantic filtering of foreground-focused augmentations improves clean generalization and adversarial robustness without adversarial-example generation during training.

A. Shortcut Learning and Foreground-Aware Augmentation

Shortcut learning arises when models rely on predictive but non-causal features, such as background, texture, or acquisition artifacts. Makar et al. used auxiliary labels to suppress known shortcut attributes through causal supervision [1]. However, such methods require prior knowledge or annotation of the spurious features. Our approach instead reduces background influence and filters transformed samples without explicit shortcut labels.

Foreground-aware augmentation provides greater control than whole-image transformations. ObjectAug applies transformations directly to segmented objects [9], while Mask2Former supports general-purpose foreground localization through masked attention [10]. These studies motivate our use of pretrained localization models, but they do not examine whether human validation of augmented samples improves classification robustness.

B. Adversarially Robust Representation Learning

Self-supervised and semi-supervised methods have been widely studied as alternatives or complements to fully supervised adversarial training. Alayrac et al., Goyal et al., Naseer et al., and Li et al. showed that unlabeled or partially labeled data can improve robust representation learning [2]–[5]. Adversarial contrastive learning further improves downstream robustness by incorporating perturbations into representation pretraining [6], [7]. These methods are effective but generally require adversarial-example generation or specialized robust optimization.

Related approaches have addressed limited-data and domain-specific settings. Xu et al. studied adversarial self-supervised learning for SAR recognition [14], Dong et al. proposed robust few-shot learning through parameter co-distillation [8], and Shim and Jo combined denoising and identity preservation for medical-image robustness [11]. These

studies motivate auxiliary representation-preservation objectives, whereas our work focuses on the quality and selection of augmented training samples.

C. Augmentation and Data-Centric Defenses

Carefully designed augmentation can improve invariance and robustness. Huh et al. demonstrated this principle in self-supervised and unsupervised speaker recognition [15], [16]. Other approaches modify inputs at inference time; for example, Wu et al. proposed runtime masking and cleansing to suppress adversarial regions [17]. Our method instead intervenes during training by removing semantically invalid augmentations before optimization.

Contemporary data-centric AI frameworks emphasize improving training data through systematic selection, debugging, and validation rather than modifying model architectures alone [18]. Representative-subset methods, such as core-set selection, retain samples that provide broad coverage of the learned feature space [19]. More recently, vision–language models such as CLIP have enabled semantic filtering through image–text similarity [20], and large-scale frameworks such as DataComp have demonstrated the effectiveness of CLIP-based filtering for multimodal dataset construction [21].

These approaches address objectives related but not identical to ours. Core-set selection primarily optimizes representativeness and diversity, whereas CLIP-based filtering evaluates image–text alignment. Our filtering stage instead determines whether an augmentation preserves the assigned class and class-defining foreground structure. The automated condition provides a confidence-based semantic-filtering baseline, while the human condition examines whether direct visual judgment contributes additional value. Comparisons with core-set and vision–language filtering constitute an important direction for future work.

A limitation of standard augmentation is that all generated samples are typically accepted automatically. Aggressive cropping, erasing, or color transformations may remove class-defining information or create ambiguous samples. Confidence-based filtering can reject candidates inconsistent with their labels, but its decisions may reflect the same shortcuts or calibration errors as the classifier. Human review provides an independent semantic signal, although it introduces cost and subjectivity.

D. Transfer Learning and Research Gap

Robustness also depends on architecture and pretraining. Rando et al. examined adversarial behavior in DINO-based Vision Transformers [22], while Vaishnavi et al. showed that pretraining can influence downstream robustness [12]. RobustArt further demonstrated that robustness conclusions may vary across architectures and training configurations [13]. These findings motivate our evaluation with both a custom CNN and four pretrained encoders: MobileNetV2, ResNet50, VGG16, and VGG19.

Prior work has established the value of shortcut supervision, foreground localization, robust pretraining, and auxiliary

reconstruction objectives. However, limited attention has been given to the quality-control stage applied after augmentation. In particular, it remains unclear whether human filtering provides benefits beyond confidence-based filtering and whether improved clean generalization consistently translates into adversarial robustness. We address this gap through a controlled comparison of unfiltered, automated, and human-supervised augmentation across datasets, attacks, and model architectures.

III. METHODOLOGY

A. Framework Overview

Let $\mathcal{D} = (x_i, y_i)_{i=1}^N$ denote a labeled image dataset, where x_i is an input image and y_i is its class label. The proposed framework consists of four stages: foreground-focused preprocessing, candidate augmentation, augmentation-quality filtering, and dual-objective model training. As illustrated in Fig. 1, each source image is first processed to reduce potentially distracting background information. Multiple transformed views are then generated from the processed image and collected into a common candidate pool. This same pool is used to construct three training conditions: unfiltered augmentation, confidence-based automated filtering, and human-supervised filtering.

The framework is designed to investigate whether semantic quality control improves model generalization and whether human judgment provides benefits beyond automated filtering. It does not generate adversarial examples during training and is therefore not a form of adversarial training. Instead, it constitutes a data-centric training strategy whose robustness is evaluated after model optimization.

B. Foreground-Focused Preprocessing

For each image x_i , the preprocessing operator $\mathcal{S}_d(\cdot)$ produces a foreground-focused reference image $r_i = \mathcal{S}_d(x_i)$, where the operator depends on dataset d . The objective is not to obtain a complete semantic segmentation of every object, but to reduce the influence of background regions while preserving class-relevant foreground content.

MNIST images contain a nearly uniform background and therefore do not require a natural-image segmentation model. A binary foreground mask is obtained directly from pixel intensity, resized to the original 28×28 resolution when necessary, and applied to the grayscale source image. This deterministic procedure avoids the domain mismatch that would result from applying a natural-image segmenter pretrained on ImageNet- or COCO-style categories to handwritten digits.

For CIFAR-10 and STL-10, foreground localization is derived from the attention maps of a pretrained DINO ViT-S/8 encoder. Each image is resized to 224×224 , and the class-token attention assigned to the patch tokens in the final transformer block is averaged across attention heads. The resulting attention map is normalized, reshaped to its spatial patch arrangement, and thresholded at $\tau_m = 0.1$ to obtain a binary localization mask m_i . The mask is then resized to the native image resolution.

Because low-resolution attention maps may imperfectly localize object boundaries, the natural-image datasets use

background blurring rather than hard removal. The foreground-focused reference is computed as

$$r_i = m_i \odot x_i + (1 - m_i) \odot G_{\sigma_d}(x_i), \quad (1)$$

where $\odot$ denotes element-wise multiplication and G_{σ_d} is Gaussian blurring with $\sigma_d = 2$ for CIFAR-10 and $\sigma_d = 3$ for STL-10. This operation retains the complete image while reducing high-frequency background detail. All reference images and localization masks are computed once and cached to avoid repeated execution of the pretrained encoder during model training.

C. Candidate Augmentation

For each reference image r_i , a transformation $a_{i,j}$ is sampled from a dataset-specific augmentation distribution $\mathcal{A}_d$. The j th candidate associated with image i is

$$\tilde{x} * i, j = a * i, j(r_i), \quad j \in 1, \dots, M_d, \quad (2)$$

where M_d is the number of augmented views generated per source image. The complete candidate pool is therefore

$$\mathcal{C} = \{(\tilde{x} * i, j, r_i, y_i)\} * i = 1, j = 1^{N, M_d}. \quad (3)$$

For MNIST, $M_d = 2$, and the augmentation distribution contains random cropping with reflection padding, brightness variation, and random erasing. Horizontal flipping is excluded because it is not reliably label preserving for handwritten digits. For CIFAR-10 and STL-10, $M_d = 3$, and the augmentation distribution additionally includes random horizontal flipping, rotation within $\pm 30^\circ$, stronger random cropping, per-channel color jitter, grayscale conversion, and random erasing. The augmentation parameters are sampled once to construct a shared candidate pool; consequently, differences among the three training conditions arise from candidate selection rather than from different randomly generated transformations.

D. Augmentation-Filtering Conditions

1) *Unfiltered Augmentation*: The unfiltered condition retains candidates without applying a semantic quality gate $\mathcal{C}_U = \mathcal{C}$. This condition measures the effect of applying the complete augmentation pipeline without inspecting whether each generated sample remains semantically valid.

2) *Confidence-Based Automated Filtering*: The automated condition uses an auxiliary classifier $q_\psi(y \mid \tilde{x})$ to assess whether a candidate remains consistent with its assigned class. A candidate is accepted when

$$\arg \max_c q_\psi(c \mid \tilde{x} * i, j) = y_i \quad \text{and} \quad q * \psi(y_i \mid \tilde{x}_{i,j}) \geq \tau_c, \quad (4)$$

where $\tau_c = 0.70$ is fixed before test evaluation.

To prevent circular filtering, q_ψ is trained independently of the candidate samples it evaluates. Specifically, the auxiliary classifier is trained only on the original training partition, or its predictions are obtained through cross-fitting such that each candidate is scored by a model that was not trained on that candidate. The auxiliary classifier is used only as a quality gate

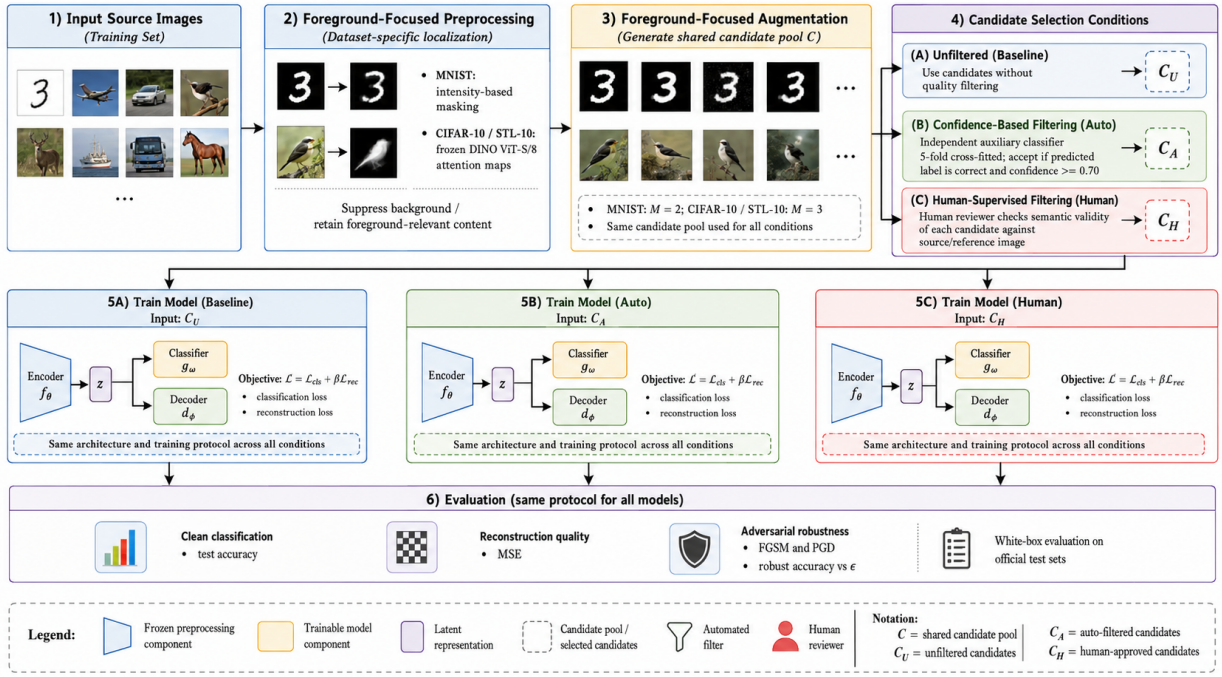


Fig. 1. Overview of the proposed framework. Foreground-focused source images are augmented to form a shared candidate pool. The candidates are then used without filtering, filtered by an independent confidence-based classifier, or reviewed by a human evaluator. Models trained under the three conditions are evaluated on clean classification and adversarial robustness.

and is not reused as the final automatically filtered model. The accepted set is denoted by

$$\mathcal{C} * \mathcal{A} = \{(\tilde{x} * i, j, r_i, y_i) \in \mathcal{C} : \text{Eq. (4) holds}\}. \quad (5)$$

3) *Human-Supervised Filtering*: For the human-supervised condition, each candidate is displayed beside its corresponding reference image and class label through an interactive review interface. The reviewer assigns a binary accept-or-reject decision according to three criteria:

- 1) the transformed image remains consistent with the assigned class;
- 2) the class-defining foreground structure remains visible and identifiable; and
- 3) the transformation does not introduce severe cropping, erasing, color distortion, or artifacts that alter the sample’s semantic interpretation.

The human-approved candidate set is

$$\mathcal{C} * \mathcal{H} = \{(\tilde{x} * i, j, r_i, y_i) \in \mathcal{C} : h(\tilde{x}_{i,j}, r_i, y_i) = 1\}, \quad (6)$$

where $h(\cdot) \in [0, 1]$ denotes the reviewer’s decision. The human gate is intended to reject semantically invalid transformations; it is not assumed to identify every shortcut or guarantee adversarial robustness.

4) *Controlled Comparison*: Filtering changes both sample quality and the number of retained candidates. To separate these effects, the primary comparison uses class- and sample-matched subsets of $\mathcal{C} * \mathcal{U}$, $\mathcal{C} * \mathcal{A}$, and $\mathcal{C}_H$. For each class,

the common sample count is set to the smallest number available across the filtered conditions. Candidates are then selected without replacement, with the unfiltered condition using a random subset of the shared pool. All three models consequently receive the same number of samples per class and the same optimization budget. Acceptance rates and results obtained using the complete filtered sets may be reported separately as a practical-yield analysis.

E. Dual-Head Model Architecture

The trainable model contains a shared encoder f_θ , a classification head g_ω , and a reconstruction decoder d_ϕ . Given an augmented candidate $\tilde{x}_{i,j}$, the encoder produces a latent representation

$$z_{i,j} = f_\theta(\tilde{x}_{i,j}) \in \mathbb{R}^{128}. \quad (7)$$

The classification head produces a probability distribution over the K target classes:

$$\hat{y} * i, j = g * \omega(z_{i,j}) = \text{softmax}(g_\omega(z_{i,j})). \quad (8)$$

The decoder reconstructs the foreground-focused source image:

$$\hat{r} * i, j = d * \phi(z_{i,j}). \quad (9)$$

For MNIST and CIFAR-10, the encoder contains two convolutional blocks. Each block contains two convolutional layers followed by batch normalization and ReLU activation, with

max pooling applied after the block. MNIST uses a single-channel 28×28 input, whereas CIFAR-10 uses a three-channel 32×32 input. For STL-10, a third convolutional block is added to accommodate the 96×96 input resolution. The corresponding decoder uses transposed-convolution layers to reconstruct an image at the native dataset resolution. The architecture is identical across filtering conditions within each dataset.

F. Dual-Objective Training Loss

The classification component is the average cross-entropy loss over a mini-batch $\mathcal{B}$:

$$\mathcal{L} * \text{cls} = -\frac{1}{|\mathcal{B}|} \sum_{*(i,j) \in \mathcal{B}} \sum_{c=1}^K y_{i,c} \log \hat{y}_{i,j,c}, \quad (10)$$

where $y_{i,c}$ is the one-hot class indicator and $\hat{y}_{i,j,c}$ is the predicted probability for class c .

The reconstruction component measures the normalized mean-squared error between the decoded output and the foreground-focused source image:

$$\mathcal{L} * \text{rec} = \frac{1}{|\mathcal{B}|CHW} \sum_{*(i,j) \in \mathcal{B}} |\hat{r}_{i,j} - r_{i,j}|_2^2, \quad (11)$$

where C , H , and W denote the number of channels, image height, and image width, respectively. Normalization by the number of pixels makes the reconstruction term comparable across image resolutions and eliminates the need for dataset-specific scaling by $H \times W$.

The complete objective is $\mathcal{L} = \mathcal{L} * \text{cls} + \beta \mathcal{L} * \text{rec}$, where $\beta \geq 0$ controls the contribution of reconstruction. The reconstruction objective encourages different augmented views of the same source image to retain sufficient information to recover their common reference image. It therefore promotes source-level consistency without treating individual image identities as additional classification labels.

The value of β is selected using a held-out validation partition and is fixed before evaluating the test set. Following validation, $\beta = 0.5$ is used for the primary experiments. Test-set performance is not used to select this value.

G. Dataset-Specific Architectural Adaptations

For STL-10, max pooling across three convolutional blocks reduces the spatial resolution from 96×96 to 12×12 before projection into the 128-dimensional latent representation. The decoder correspondingly contains three upsampling or transposed-convolution stages to reconstruct the 96×96 reference image. CIFAR-10 and STL-10 use the DINO-based foreground-localization procedure in Eq. (1), whereas MNIST uses intensity-based foreground extraction. Apart from these resolution- and modality-specific adaptations, the filtering conditions within each dataset use identical preprocessing, model architecture, loss function, training schedule, and candidate-generation process.

IV. EXPERIMENTAL SETUP

Datasets and conditions. We evaluate three training conditions on MNIST, CIFAR-10, and STL-10: unfiltered augmentation (Baseline), confidence-based automated filtering (Auto), and human-supervised filtering (Human). The official test sets are used only for final evaluation, while 10% of each training set is reserved for validation. Experiments are repeated with five random seeds, 0, 1, 2, 3, 4, and results are reported as mean $\pm$ standard deviation.

For MNIST, 1,000 source images are augmented with $M = 2$ views, producing 2,000 candidates; 1,876 are human-approved. For CIFAR-10, 3,000 sources and $M = 3$ produce 9,000 candidates, of which 4,690 are human-approved and balanced to 469 per class. For STL-10, 1,000 sources and $M = 3$ produce 3,000 candidates; 2,683 are human-approved and 2,590 pass the automated filter. To isolate filtering quality from dataset size, the primary comparison uses class-matched subsets with equal candidate counts across all three conditions.

Automated filtering. A candidate is accepted when an auxiliary classifier predicts its assigned class with confidence at least 0.70. To avoid circular filtering, five-fold cross-fitting is used: candidates from each fold are scored by a classifier trained only on the other four folds. The auxiliary models are discarded after filtering and are not used to initialize the final models.

Architecture. The custom model consists of a convolutional encoder, a 128-dimensional latent representation, a classification head, and a reconstruction decoder. Each encoder block contains two 3×3 convolutional layers with stride 1, batch normalization, ReLU activation, and 2×2 max pooling. MNIST and CIFAR-10 use two blocks with channel widths 32 and 64, whereas STL-10 uses three blocks with widths 32, 64, and 128. The decoder uses 2 transposed-convolution layers for MNIST and CIFAR-10, and 3 for STL-10, with 4×4 kernels and stride 2. The models contain 939,275 trainable parameters for MNIST, 1,188,109 for CIFAR-10, and 5,293,069 for STL-10.

Training. All models are trained for 40 epochs with batch size 64 and 300 steps per epoch using the Adam optimizer. MNIST uses a fixed learning rate of 10^{-3} . CIFAR-10 and STL-10 use a five-epoch linear warmup to 10^{-3} followed by cosine decay. Each batch contains equal numbers of foreground-focused source images and condition-specific augmented candidates. The reconstruction weight is selected on the validation set and fixed at $\beta = 0.5$ for the primary experiments.

Foreground preprocessing. MNIST foregrounds are extracted using intensity-based masking. CIFAR-10 and STL-10 use frozen DINO ViT-S/8 attention maps. Images are resized to 224×224 , attention is averaged across heads and thresholded at 0.1, and background regions are blurred using $\sigma = 2$ for CIFAR-10 and $\sigma = 3$ for STL-10. All processed images are cached before training.

Adversarial evaluation. We evaluate untargeted white-box FGSM and PGD attacks in the classifier-input space. For each

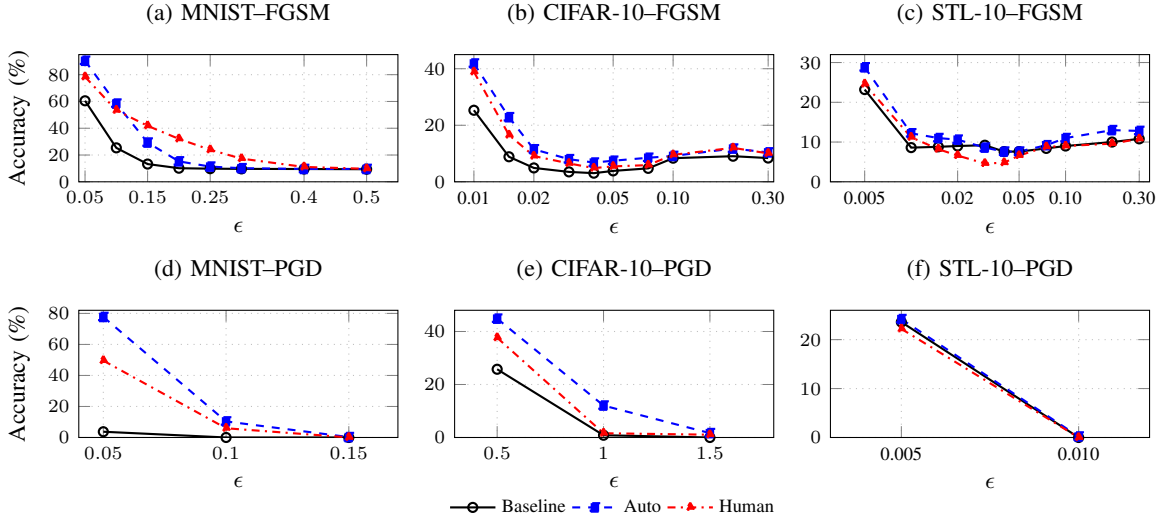


Fig. 2. Adversarial accuracy under FGSM and PGD. Auto generally provides stronger low-perturbation robustness, while Human becomes stronger under moderate FGSM perturbations on MNIST. PGD accuracy approaches zero at larger perturbation budgets. Clean accuracy at $\epsilon = 0$ is reported separately in Fig. 3.

test image x_i , the fixed preprocessing operator is first applied to obtain $r_i = S_d(x_i)$. The adversarial perturbation is then optimized directly with respect to r_i using $\nabla_{r_i} \mathcal{L}_{\text{cls}}(F_{\theta, \omega}(r_i), y_i)$, where $F_{\theta, \omega} = g_{\omega} \circ f_{\theta}$ denotes the trained encoder–classifier composition. The perturbation is constrained by an ℓ_{∞} budget, and adversarial inputs are clipped to $[0, 1]$.

The DINO attention extraction and binary-thresholding operations are not included in the attack graph because foreground-focused images are generated and cached as an offline data-preparation step rather than used as an inference-time defense. Accordingly, the reported experiments measure robustness to perturbations of the foreground-focused classifier inputs and do not claim end-to-end robustness against perturbations of the original raw images.

PGD uses 40 iterations, a step size of $\alpha = \epsilon/10$, random initialization, and ten restarts. MNIST is evaluated at $\epsilon \in \{0, 0.05, \dots, 0.50\}$, whereas CIFAR-10 and STL-10 use $\epsilon \in \{0, 0.005, \dots, 0.30\}$. STL-10 evaluation uses batches of 500 samples to satisfy GPU-memory constraints.

Transfer learning. MobileNetV2, ResNet50, VGG16, and VGG19 pretrained on ImageNet are evaluated on both CIFAR-10 and STL-10. For MobileNetV2 and ResNet50, the last 35 layers are fine-tuned; for VGG16 and VGG19, the last 4 layers are fine-tuned. The classification and reconstruction heads are attached to the global-average-pooled representation. MobileNetV2 inputs are normalized to $[-1, 1]$; ResNet50 and VGG inputs use channel-wise zero-centering consistent with their respective ImageNet preprocessing.

Transfer models use the same training schedule, random seeds, and matched candidate counts as the custom-model experiments. Because the transfer-learning analysis is intended as a focused examination of whether the effect of human filtering generalizes across pretrained encoders, it compares the two endpoint conditions, Baseline and Human. The complete

three-way comparison among Baseline, Auto, and Human is retained for the primary custom-architecture experiments. All experiments are implemented using Tensorflow 2.20 and executed on an Nvidia Tesla T4 GPU.

V. RESULTS

Fig. 3 compares clean test accuracy across the three datasets. Human filtering achieves the highest accuracy with the custom architecture in all cases, although the gain is marginal on MNIST and more pronounced on CIFAR-10 and STL-10.

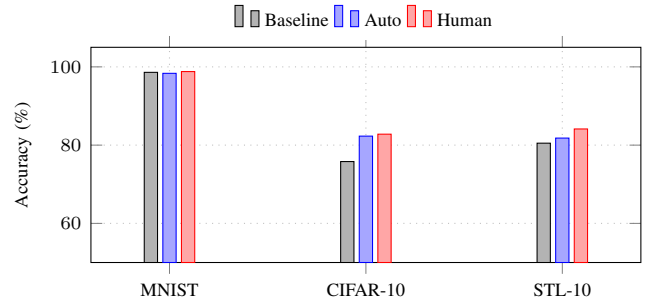


Fig. 3. Clean test accuracy across datasets and training conditions.

A. MNIST

As shown in Table I, all conditions achieve comparable clean accuracy. Human reaches 98.80%, compared with 98.60% for Baseline and 98.35% for Auto. Its slightly higher reconstruction MSE indicates that the classification gain does not result from improved reconstruction.

Fig. 2 summarizes FGSM and PGD performance across the three datasets. Under FGSM, Auto performs best at low perturbation levels, whereas Human becomes stronger at $\epsilon = 0.15$ – 0.30 . At $\epsilon = 0.20$, Human achieves 32.05%, compared with

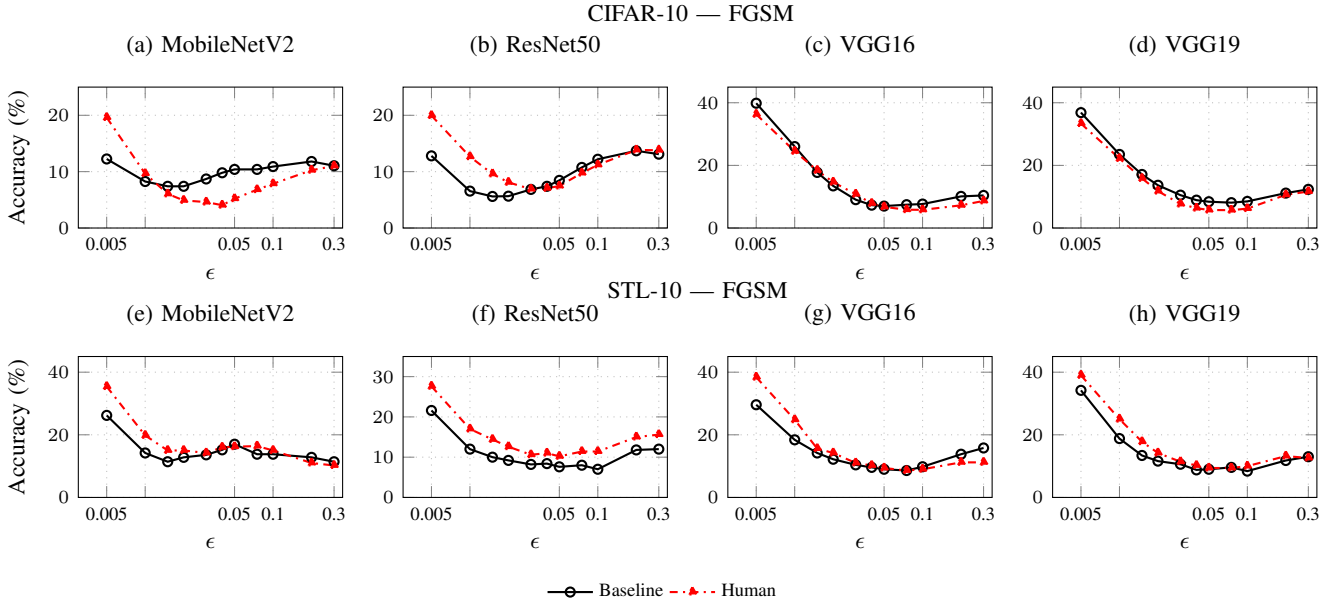


Fig. 4. FGSM adversarial accuracy under transfer learning on CIFAR-10 (top row, panels a–d) and STL-10 (bottom row, panels e–h). Human filtering improves robustness at low ϵ for MobileNetV2 and ResNet50 on both datasets. For VGG16 and VGG19 on CIFAR-10, Baseline achieves higher robustness at most perturbation levels, consistent with the clean accuracy reversal. Clean accuracy at $\epsilon = 0$ is reported separately in Fig. 3.

TABLE I
MNIST CLEAN PERFORMANCE.

Metric	Baseline	Auto	Human
Accuracy (%)	98.60	98.35	98.80
Reconstruction MSE	0.01672	0.01744	0.01875

15.25% for Auto. Under PGD, Auto performs best at $\epsilon = 0.05$, but all conditions deteriorate rapidly as ϵ increases.

B. CIFAR-10

Table II shows that Human achieves the highest clean accuracy at 82.80%, followed closely by Auto at 82.30%. Both substantially outperform Baseline, while reconstruction errors remain similar.

TABLE II
CIFAR-10 CLEAN PERFORMANCE.

Metric	Baseline	Auto	Human
Accuracy (%)	75.80	82.30	82.80
Reconstruction MSE	0.02726	0.02753	0.02640

Under FGSM, both filtered models outperform Baseline at most perturbation levels, with Auto generally performing best at low and moderate ϵ values. At $\epsilon = 0.010$, Auto, Human, and Baseline achieve 41.85%, 38.90%, and 25.30%, respectively. Under PGD, Auto also provides the strongest low- ϵ robustness, but all conditions collapse at $\epsilon \geq 0.020$.

C. Transfer Learning

Table III summarizes clean accuracy for all four pretrained encoders on CIFAR-10 and STL-10. On CIFAR-10, human filtering improves clean accuracy for MobileNetV2 (+8.20 pp)

and ResNet50 (+0.80 pp), but reduces accuracy for VGG16 (-8.80 pp) and VGG19 (-10.20 pp). On STL-10, human filtering consistently improves clean accuracy across all four architectures: MobileNetV2 (+4.20 pp), ResNet50 (+2.35 pp), VGG16 (+6.90 pp), and VGG19 (+4.45 pp).

The reversal observed for VGG16 and VGG19 on CIFAR-10 suggests that the benefit of human-curated augmentation diversity is modulated by both architecture and resolution. VGG architectures lack residual connections and may be more sensitive to distribution shifts introduced by diverse augmentations at 32×32 , whereas the same architectures benefit from human filtering at 96×96 .

As shown in Fig. 4, human filtering improves low- ϵ FGSM robustness for MobileNetV2 and ResNet50 on both CIFAR-10 and STL-10. For VGG16 and VGG19 on CIFAR-10, the Baseline models achieve higher adversarial accuracy at most perturbation levels, consistent with their higher clean accuracies. On STL-10, Human provides stronger low- ϵ robustness across all four pretrained encoders. Under PGD, all transfer-learning conditions deteriorate rapidly, consistent with the custom-model findings.

TABLE III
TRANSFER LEARNING CLEAN ACCURACY (%).

Architecture	Dataset	Baseline	Human
MobileNetV2	CIFAR-10	51.70	59.90
ResNet50	CIFAR-10	52.10	52.90
VGG16	CIFAR-10	65.35	56.55
VGG19	CIFAR-10	63.05	52.85
MobileNetV2	STL-10	72.10	77.85
ResNet50	STL-10	83.65	86.00
VGG16	STL-10	72.05	78.95
VGG19	STL-10	74.50	78.95

D. STL-10

On STL-10, Human achieves 84.13% clean accuracy, exceeding Auto by 2.33 percentage points and Baseline by 3.63 points (Table IV). However, Human also produces the highest reconstruction MSE, showing that improved classification does not necessarily coincide with better reconstruction.

TABLE IV
STL-10 CLEAN PERFORMANCE.

Metric	Baseline	Auto	Human
Accuracy (%)	80.50	81.80	84.13
Reconstruction MSE	0.03342	0.03675	0.04013

Under FGSM, Auto generally achieves the highest adversarial accuracy, including 28.80% at $\epsilon = 0.005$, compared with 24.60% for Human and 23.20% for Baseline. Under PGD, the differences are small at $\epsilon = 0.005$, and all models collapse at larger perturbations.

Overall, Human consistently provides the highest clean accuracy with the custom architecture, while Auto generally provides greater low-strength adversarial robustness on the natural-image datasets. The relative robustness benefit of human filtering is therefore dataset- and attack-dependent, and neither filtering strategy replaces dedicated adversarial training.

VI. CONCLUSION

This paper presented a human-guided augmentation framework combining foreground-focused preprocessing, systematic augmentation, semantic filtering, and dual-objective classification–reconstruction learning. On MNIST, CIFAR-10, and STL-10, human filtering achieved the highest clean accuracy with the custom architecture, improving over the unfiltered baseline by 0.20, 7.00, and 3.63 percentage points, respectively. Its adversarial benefit was dataset dependent: it outperformed automated filtering under moderate-to-high FGSM perturbations on MNIST, whereas confidence-based filtering generally matched or exceeded it on CIFAR-10 and STL-10. All methods degraded substantially under stronger PGD attacks, showing that augmentation filtering complements rather than replaces adversarial training. Under transfer learning, human filtering improved clean accuracy across all four pretrained encoders on STL-10 but reduced it for VGG16 and VGG19 on CIFAR-10, indicating sensitivity to encoder architecture and input resolution. Limitations include ten-class datasets, one human reviewer, and white-box attacks. Future work will consider larger and fine-grained datasets, multiple reviewers, black-box and transfer attacks, and vision–language models for scalable quality control.

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